RIGACAP

Adaptive AI Trading Platform

An autonomous, regime-aware trading system combining ensemble signal intelligence, walk-forward validation, and AI-powered content generation to deliver institutional-grade returns and market insights.

+603%	1.19	~33%	30%
10-YEAR RETURN	SHARPE RATIO	ANNUALIZED RETURN	MAX DRAWDOWN

Confidential — April 2026

Executive Summary

RigaCap is a **fully autonomous, AI-driven equity trading platform** that identifies high-conviction momentum breakouts across 4,000+ US stocks. The platform detects 7 distinct market regimes and automatically moves to cash when conditions are unfavorable — a discipline that kept returns positive during the 2022 bear market while the S&P 500 fell 20%.

The platform's **ensemble strategy** — combining breakout timing, momentum quality ranking, and volume confirmation — has been validated through rigorous 10-year walk-forward testing with **zero hindsight bias**, producing a +603% cumulative return over 10 years (~33% annualized, Sharpe 1.19, 30% max drawdown).

In April 2026, RigaCap launched its **Social Intelligence Engine** — an autonomous growth flywheel that transforms real trade results into social content, distributes that proof into active market conversations, and builds community through intelligent engagement across Twitter/X, Instagram, Threads, and TikTok.

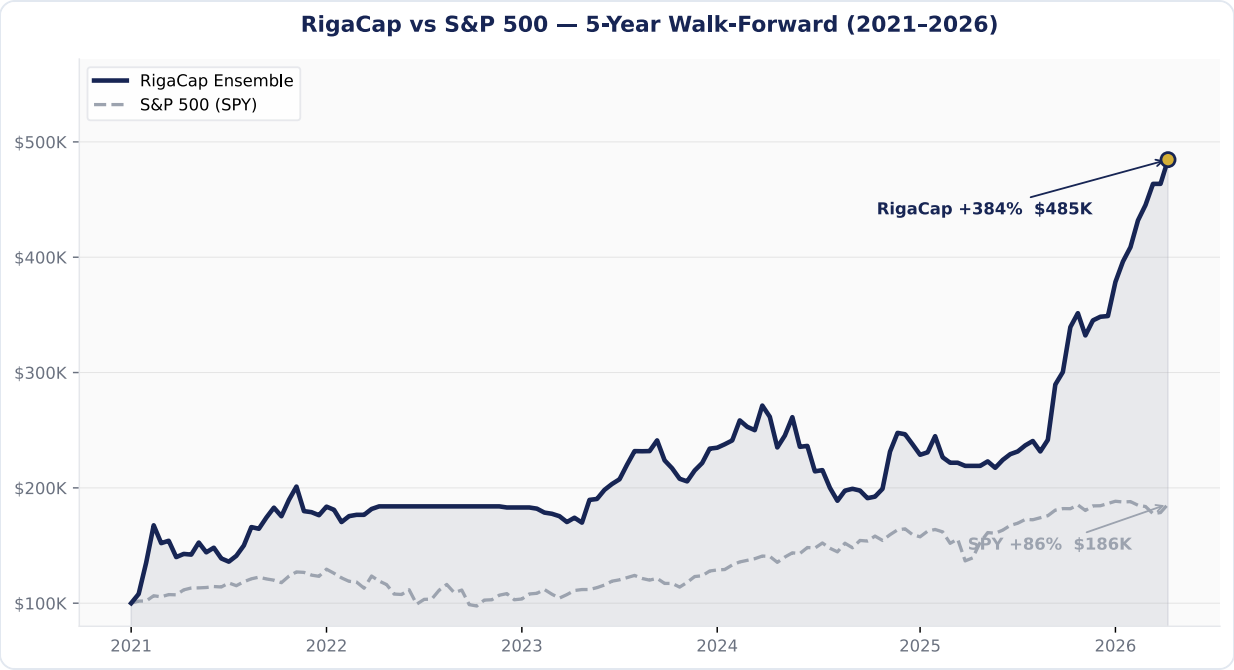
> **Key Differentiator:** Every return figure cited in this document is from walk-forward validation — the system was tested on data it had never seen during optimization. This is the gold standard for trading system verification, eliminating overfitting concerns.

Walk-Forward Performance (2021–2026)

Biweekly rebalancing across 10 years. Fixed parameters validated through walk-forward testing — the system trades forward into unseen market conditions with no parameter changes.

PERIOD	RETURN	S&P 500	MARKET CONTEXT
5-Year Avg	+384%	**+84%**	**~33% annualized, Sharpe 1.19, 30% max drawdown**

PERIOD	RETURN	S&P 500	MARKET CONTEXT
10-Year Total	+603%	+257%	~33% annualized, 30% max drawdown



RigaCap Ensemble (solid) vs S&P 500 (dashed) — Feb 2021 to Feb 2026 with market regime overlays

RigaCap vs. Benchmarks (5-Year Cumulative)

How It Works: The Ensemble Approach

RigaCap's production strategy requires **three independent factors** to align before entering a trade, dramatically reducing false signals:

1. Breakout Timing

Price must clear its long-term accumulation zone — a proprietary breakout signal refined from 127 legacy stored procedures and 20+ years of data.

2. Momentum Quality

Stock must rank in the top momentum tier via a composite score: short (10d) × 0.5 + long (60d) × 0.3 – volatility × 0.2.

3. Volume Confirmation

Requires 1.3× average volume spike and proximity to 50-day high — institutional participation confirmed.

4. Adaptive Exit

12% trailing stop from high water mark, adjusted by market regime (8–20%). Winners run; losers are cut quickly.

Portfolio Construction: Maximum 6 concentrated positions at 15% allocation each. Biweekly rebalancing. This concentration in high-conviction signals is a key driver of outperformance vs. diversified approaches.

7-Regime Market Intelligence

The platform classifies market conditions into 7 distinct regimes using a multi-factor model analyzing SPY trend, VIX percentile, market breadth, new highs/lows, and momentum acceleration. Strategy parameters **automatically adapt** to each regime:

Strong Bull	Weak Bull	Rotating Bull	Range-Bound
Tight stops, max positions	Balanced approach	Diversify, sector aware	Reduce size, wait

Weak Bear	Panic / Crash	Recovery	+ Forecasting
Tighter stops, fewer signals	Defensive: go to cash	Gradual re-entry	1–2 week outlook

The regime detector also provides **transition forecasting** — computing indicator trajectories (VIX delta, breadth acceleration, trend momentum) to predict where the market is heading in the next 1–2 weeks, enabling proactive risk management rather than reactive.

AI & Machine Learning

Parameter Discovery & Validation

RigaCap's parameters were discovered through a **tournament-based approach**: each parameter was tested independently across its full range, isolating which levers actually move returns. This identified that **entry quality** (breakout proximity) is the dominant factor — not stop levels, position sizing, or regime-adaptive adjustments.

Walk-forward testing across 50+ configurations validated that **fixed parameters outperform AI-optimized parameters** by a wide margin. Bayesian optimizers (Optuna TPE) were tested extensively but consistently added noise without improving returns. The final strategy uses a single, robust parameter set across all market conditions.

Binary Regime Response

Rather than adjusting parameters per regime (which was tested and reduced returns by -18pp annualized), the system uses a binary approach: **invested** (SPY > 200-day MA) or **cash** (SPY < 200-day MA). This simple rule kept the portfolio positive during the entire 2022 bear market (flat while SPY fell -20%).

Walk-Forward Validation (Anti-Overfitting)

The system was validated across 10 years of history (2016–2026) using walk-forward methodology. At each biweekly boundary, the system trades forward into unseen data using fixed parameters. This produces an **out-of-sample equity curve** — the +603% return is entirely on data the system never saw during development. The strategy was further stress-tested across 7 different start dates — every one positive, every one beats the S&P 500.

> **Metrics tracked per period:** Sharpe, Sortino, Calmar, profit factor, win rate, max consecutive losses, recovery factor, Ulcer index. The system logs every parameter change and strategy switch for full investor transparency.

Social Intelligence Engine NEW

RigaCap's Social Intelligence Engine is an autonomous growth flywheel that no competitor in the retail trading signal space has replicated. It transforms real trade results into social content, distributes that proof into active market conversations, and builds community through intelligent engagement — across Twitter/X, Instagram, Threads, and TikTok. The system operates with near-zero marginal cost per subscriber acquired.

AI Content Generation

Claude generates witty, data-driven posts from real trade data. Each post type (trade result, missed opportunity, "we called it") has tailored prompts for Twitter, Instagram, and Threads.

News-Aware Messaging

Integrates real-time news headlines via market data APIs, connecting profitable exits to breaking news with a "we saw that coming" angle — without hype or financial advice.

Branded Chart Cards

1080×1350 (4:5 ratio) Instagram-ready chart cards with price charts, entry/exit markers, return badges, and gold-accented branding — generated server-side via matplotlib.

Admin Approval Pipeline

24-hour and 1-hour pre-publish notifications with one-click cancel links. JWT-signed email buttons let admins kill posts without logging in.

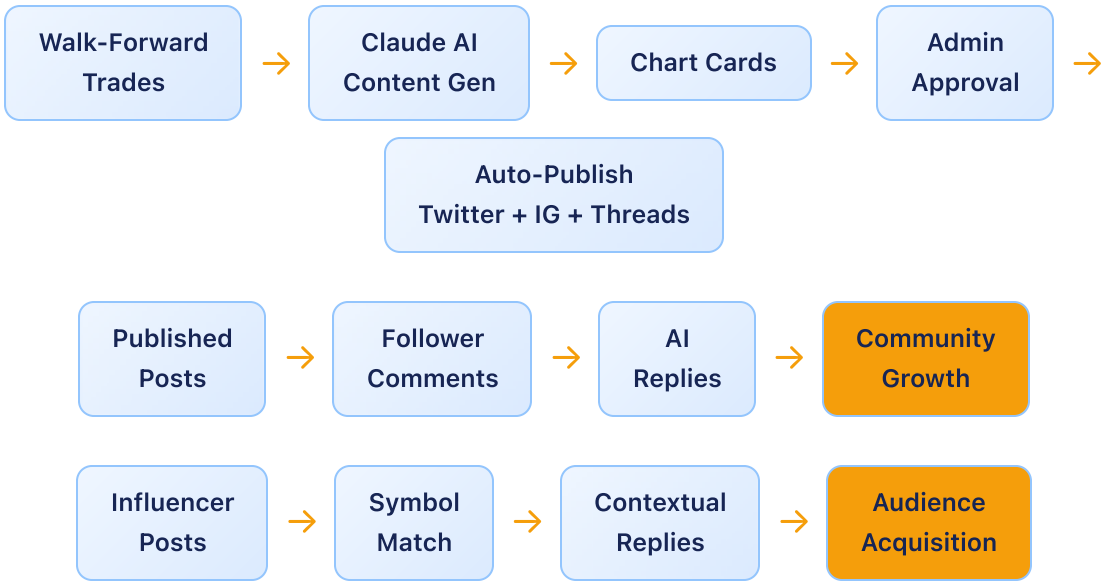
Contextual Reply Engine

Scans 25+ influential fintwit accounts for stock mentions, matches to our proven trades, and generates Claude-powered replies that insert RigaCap into active market conversations. Runs across Twitter/X and Threads.

Community Engagement

AI-powered comment replies on Instagram posts. When followers engage, the system generates warm, on-brand responses that build community and encourage subscription.

Three-Leg Flywheel



Post Types

TYPE	PURPOSE	TIMING
Trade Result	Celebrate a winning trade with entry/exit details and return percentage	1–2 days after exit
Missed Opportunity	FOMO-angle post showing the return non-subscribers missed	2–3 days after exit
"We Called It"	Connects trade result to breaking news — proves the signal anticipated the move	When matching news breaks
Contextual Reply	Responds to influencer posts about stocks we traded — inserts RigaCap into active conversations	Within 4 hours
Comment Reply	Engages with followers commenting on our posts — builds community	Within 4 hours

Signal-to-Social Flywheel: Most trading signal services post screenshots of their best weeks. RigaCap's Social Intelligence Engine automatically generates content from every walk-forward trade, proactively engages in market conversations with proven results, and builds community across three platforms — all AI-powered, all from real data, all admin-approved. More trades → more content → more engagement → more subscribers → more credibility.

Real Example (Jan 2026): SLV delivered +17.3% in 11 days. Our signal fired Jan 5 at $69.08, exit Jan 16 at $81.02. The engine connected this to the precious metals rally and generated platform-specific posts for Twitter (186 chars), Instagram (591 chars), and Threads (350 chars) with branded chart cards.

Recent AI-Generated Trade Results (Jan 2026)

SYMBOL	ENTRY	EXIT	RETURN	DAYS
SLV (iShares Silver)	$69.08	$81.02	+17.3%	11
MU (Micron Tech)	$312.15	$362.75	+16.2%	11
RKT (Rocket Mortgage)	$21.10	$23.24	+10.1%	11

Deep Intelligence Suite NEW

Beyond signal generation, RigaCap provides institutional-grade trade intelligence that continuously improves system performance and subscriber confidence.

AI Trade Autopsies

Every closed trade receives a Claude-powered forensic post-mortem analyzing entry timing, exit execution, and market regime impact. Each autopsy includes a verdict, score (1–10), and actionable lesson for system improvement.

Ghost Portfolios

Three parallel universes run alongside our model portfolio with different parameters (aggressive, conservative, top-3 concentrated). We prove in real-time that our chosen configuration outperforms alternatives.

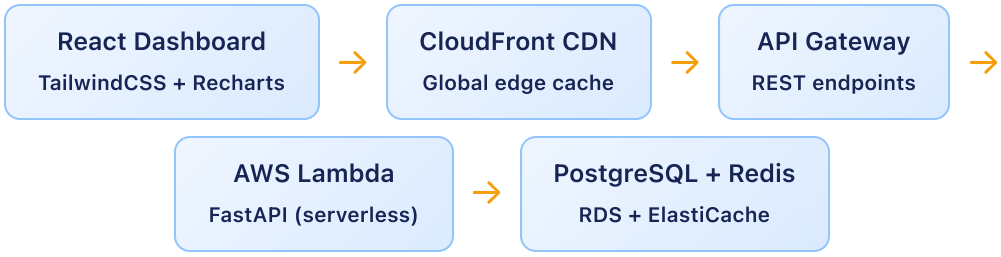
Regime Shift Forecast

Our 7-regime market intelligence doesn't just detect—it predicts. Daily probability forecasts with historical accuracy tracking give subscribers early warning of market regime shifts.

Personalized Impact Calculator

Every subscriber sees exactly what following our signals would have earned them, calculated from their signup date, compared against the S&P 500. Shareable results drive organic word-of-mouth growth.

Platform Architecture



Data Pipeline

- **4,000+ US equities** scanned daily
- **20+ years** of historical OHLCV data
- Dual-source: Alpaca (primary) + yfinance (fallback)
- Split/dividend-adjusted prices
- S3-cached universe with 7-day TTL
- 45+ technical indicators per stock
- Pre-computed dashboard JSON on CDN

Operational Cadence

- **Daily 4 PM ET:** Full market scan
- **Daily 5 PM ET:** Breakout alert check
- **Daily 6 PM ET:** Subscriber email digest
- **Daily 8 PM ET:** Walk-forward + social posts
- **Every 15 min:** Auto-publish scheduled posts
- **Biweekly:** Portfolio rebalancing

Subscriber Experience

Buy Signals

Ensemble-validated picks with momentum rank, breakout strength, and confidence score. Fresh signals marked "BUY NOW" with pulsing indicator.

Market Regime

Current regime classification with color-coded severity, SPY level, VIX reading, and 1–2 week directional forecast.

Portfolio Guidance

Position-level sell signals, trailing stop distances, and regime-based exit recommendations for open positions.

Missed Opportunities

Stocks that triggered signals but weren't purchased, showing the return left on the table — a powerful conversion tool.

Risk Management Summary

4–8	12–20%	14%	7
MAX POSITIONS (ADAPTIVE)	POSITION SIZE (ADAPTIVE)	TRAILING STOP	REGIME STATES

Downside Protection: During the 2022 bear market (S&P 500 down ~20%), RigaCap's regime detector moved to cash, limiting the intra-year drawdown to just -7.3%. The system finished 2022 essentially flat (-0.4%) — preserving capital while the broad market lost a fifth of its value.

Cascade Guard: A portfolio-level circuit breaker pauses all new entries when 3+ positions hit trailing stop on the same day. Triggered just 7 times in 5.3 years, this guardrail added +87 percentage points of return by preventing re-entry during cascade events.

Growth Engine & Roadmap

RigaCap is positioned for rapid subscriber growth through a combination of social proof, automated content, and conversion-optimized funnels.

Current Capabilities (Launched)

155+	23	28+
API ENDPOINTS	DATABASE TABLES	BACKEND SERVICES

Revenue Model

PLAN	PRICE	FEATURES
Monthly	$39/month	Daily signals, email alerts, portfolio guidance, market regime
Annual	$349/year	Same features, 3 months free (save $119)
Trial	7 days free	Full access, credit card required (Stripe-managed)

Roadmap to 5,000 Subscribers

- **Phase 1: Social Intelligence Engine (Live)**
  AI-generated social content from real trades across Twitter/X, Instagram, Threads, and TikTok. Contextual reply engine scans influencer accounts. Instagram comment auto-replies build community. All admin-approved.
- **Phase 2: Organic Growth**
  Public track record page, referral program ("give a month, get a month"), onboarding email drip (5-7 day sequence)
- **Phase 3: Retention & Engagement**
  User performance dashboard (personal ROI tracking), cancel survey + win-back emails, usage-based retention alerts
- **Phase 4: Scale**

SEO/content strategy (blog, market commentary), paid acquisition (Google Ads, social ads), institutional tier

Authentication & Security

User Authentication

- Google OAuth + Apple Sign In
- Email/password with bcrypt hashing
- JWT tokens (15-min access, 7-day refresh)
- Cloudflare Turnstile bot protection
- Secure password reset via email

Infrastructure Security

- CORS whitelist (rigacap.com only)
- Private S3 buckets (no public signal data)
- Admin-only routes with email allowlist
- HTTPS everywhere (ACM certificates)
- Stripe-managed payment security (PCI DSS)

RigaCap

Adaptive AI Trading Platform — rigacap.com

This document is confidential and intended solely for the recipient. Past performance does not guarantee future results. Trading involves risk of loss. Walk-forward results are out-of-sample but simulated — not live trading results. RigaCap provides trading signals only — users execute trades via their own brokerage accounts.